

Lecture 5: Brownian Motion & Random Lattice Walks

Leili Rafiee Sevyeri

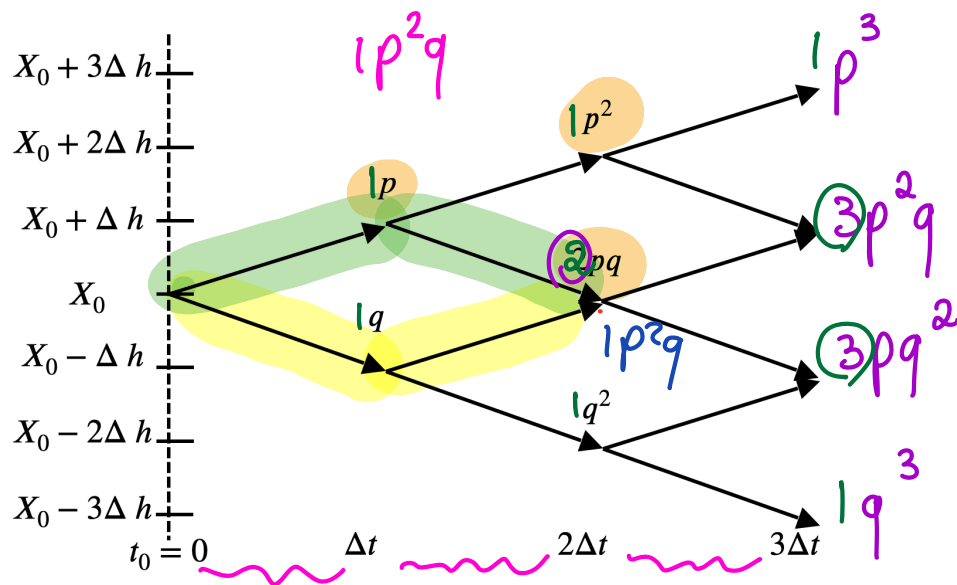
Based on lecture notes by me and many previous CS 335 instructors

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Cheriton School of Computer Science
University of Waterloo

Building the Lattice

If we repeated this for several time steps, we get a lattice structure of possible values, at each step forward in time. The probability of arriving at each lattice point is dictated by the history of steps.



End of Lecture 4

Probabilities sum to 1 at each time level. (Notice Pascal's triangle emerges in the factors!)

Lattice Model - Assumptions

We assume X follows a Markov process:

- Probability of an up/down move is independent of past moves.
- Thus, the probability distribution of future value depends only on where we are now (current value).

Some notation:

- $X_i = X(t_i)$
- $\Delta X_i = X_{i+1} - X_i$

Let's now think about the outcomes (in terms of probability distributions) resulting from such a random walk process.

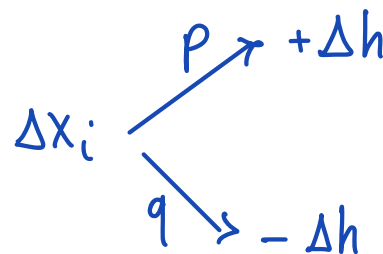
Expectation of ΔX_i (one step)

ΔX_i are independent and identically distributed (i.i.d.).

$\Delta X_i, \Delta X_j$ are pairwise independent (for $i \neq j$).

i.e., implying $E[\Delta X_i \Delta X_j] = E[\Delta X_i] E[\Delta X_j]$

A single step yields a change by ΔX_i as:



So,

$$E[\Delta X_i] = p(\Delta h) + q(-\Delta h) = (p - q)\Delta h$$

$$E[\Delta X_i^2] = p(\Delta h)^2 + q(-\Delta h)^2 = \underbrace{(p + q)}_1 (\Delta h)^2 = (\Delta h)^2$$

Variance of ΔX_i (one step)

We can then work out the variance as well:

$$\begin{aligned}\text{var}[\Delta X_i] &= E[(\Delta X_i)^2] - (E[\Delta X_i])^2 \\&= (\Delta h)^2 - (p - q)^2 (\Delta h)^2 \\&= (1 - (p - q)^2) (\Delta h)^2 \\&= (1 - (p - (1 - p))^2) (\Delta h)^2 \\&= (1 - (2p - 1)^2) (\Delta h)^2 \\&= (1 - (4p^2 - 4p + 1)) (\Delta h)^2 \\&= (-4p^2 + 4p) (\Delta h)^2 \\&= (\Delta h)^2 4p(1 - p) \\&= (\Delta h)^2 4pq\end{aligned}$$

Lattice Outcomes

Now, consider computing the result of a random walk on the lattice up to a specific finite end time t . The total number of moves/steps needed is $n = \frac{t}{\Delta t}$.

Looking at the lattice (lecture 4, slide 13), we observe that after n steps, the probability of a given lattice point satisfies:

$$p[\text{j up moves, } n\text{-j down moves}] = \frac{n!}{j!(n-j)!} p^j q^{n-j} = \binom{n}{j} p^j q^{n-j}$$

↙
"notation for
"n choose j"

which is a binomial distribution.

Expectation of $X_n - X_0$ (many steps)

The expectation of the *total* change over many steps is:

$$\begin{aligned} E[X_n - X_0] &= E\left[\sum_{i=0}^{n-1} \Delta X_i\right] \\ &= \sum_{i=0}^{n-1} E[\Delta X_i] \\ &= \sum_{i=0}^{n-1} (p-q) \Delta h \\ &= n (p-q) \Delta h \\ &= \frac{t}{\Delta t} (p-q) \Delta h \end{aligned}$$

Variance of $X_n - X_0$ (many steps)

Similarly, the variance after many steps can be shown to be:

$$\begin{aligned} \text{var}[X_n - X_0] &= \text{var} \left[\sum_{i=0}^{n-1} \Delta X_i \right] \\ &= \dots \text{???} \\ &= \frac{t}{\Delta t} 4pq(\Delta h)^2 \end{aligned}$$

Variance of $X_n - X_0$ (many steps)

$$\begin{aligned}\text{Var}[X_n - X_0] &= \text{Var}\left[\sum_{i=0}^{n-1} \Delta X_i\right] \\&= E\left[\left(\sum_i \Delta X_i - E\left[\sum_i \Delta X_i\right]\right)^2\right] \quad (\text{by def. of Variance}) \\&= E\left[\left(\sum_i \Delta X_i - \sum_i E[\Delta X_i]\right)^2\right] \quad (\text{by linearity of } E) \\&= E\left[\left(\sum_i (\Delta X_i - E[\Delta X_i])\right)^2\right]\end{aligned}$$

Define : $a_i = \Delta X_i - E[\Delta X_i]$

Then :

$$= E\left[\left(\sum_i a_i\right)^2\right] = E\left[\left(\sum_i a_i\right)\left(\sum_j a_j\right)\right]$$

Variance of $X_n - X_0$ (many steps)

$$\begin{aligned} &= E \left[\sum_{i,j} a_i a_j \right] \\ &= E \left[\sum_{i=j} a_i^2 + \sum_{i \neq j} a_i a_j \right] \\ &= \sum_i E[a_i^2] + \underbrace{\sum_{i \neq j} E[a_i a_j]}_{=0} \end{aligned}$$

Note that:

$$E[a_i a_j] = E[(\Delta x_i - E[\Delta x_i])(\Delta x_j - E[\Delta x_j])].$$

Since Δx_i and Δx_j are independent:

Variance of $X_n - X_0$ (many steps)

$$\begin{aligned} &= E[\Delta X_i - E[\Delta X_i]] \cdot E[\Delta X_j - E[\Delta X_j]] \\ &= (E[\Delta X_i] - E[\Delta X_i]) \cdot (E[\Delta X_j] - E[\Delta X_j]) \\ &= 0 \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Var}[X_n - X_0] &= \sum_i E[a_i^2] = \sum_i E[(\Delta X_i - E[\Delta X_i])^2] \\ &= \sum_i \text{Var}[\Delta X_i] \quad (\text{by def. of Variance}) \\ &= \sum_{i=0}^{n-1} 4pq(\Delta h)^2 = n 4pq(\Delta h)^2 \\ &= \frac{t}{\Delta t} 4pq(\Delta h)^2 \end{aligned}$$

Limiting Behaviour of Lattice Model/Random Walk

Why are we doing this?

We hope to recover the behaviour of the true SDE as the limit of this discrete lattice process, as the discrete time step $\Delta t \rightarrow 0$; this will enable us to use this discrete lattice walk to approximate Brownian motion.

How must we choose the increment Δh and probabilities p and q to match the SDE behaviour?

We will let $\Delta t \rightarrow 0$, $n \rightarrow \infty$, and $n\Delta t = T = \text{constant}$.

Assume $\lim_{\Delta t \rightarrow 0} pq \neq 0$ (since otherwise, this is not a random walk!)

Choice of Δh

We saw that $\text{var}[X_n - X_0] = \frac{t}{\Delta t} 4pq(\Delta h)^2$.

If $\text{var}[X_n - X_0] = 0$ or ∞ as $\Delta t \rightarrow 0$, this would not be a useful case (stocks don't exhibit either of these properties!)

To avoid these cases, we require $\Delta h^2 \propto \Delta t$, to cancel out the $1/\Delta t$ in the variance.

Therefore, $\Delta h = C_1 \sqrt{\Delta t}$, where C_1 is a constant independent of Δt .

Choice of p and q

Plugging $\Delta h = C_1 \sqrt{\Delta t}$ back into our expectation and variance expressions...

$$E[X_n - X_0] = \frac{t}{\Delta t} (p - q) \Delta h = \frac{t}{\Delta t} (p - q) C_1 \sqrt{\Delta t} = \frac{C_1 t (p - q)}{\sqrt{\Delta t}}$$

$$\text{var}[X_n - X_0] = \frac{t}{\Delta t} 4pq (\Delta h)^2 = \frac{t}{\Delta t} 4pq C_1^2 \cancel{\Delta t} = 4pq t C_1^2$$

No dependence on Δt now.
Great!

Now, apply the same logic to $E[X_n - X_0]$ above - it must also be bounded as $\Delta t \rightarrow 0$, so we want the Δt to cancel:

$$p - q \propto \sqrt{\Delta t} \rightarrow p - q = C_2 \sqrt{\Delta t} \text{ with } C_2 \text{ a constant independent of } \Delta t.$$

$$\text{Hence, } E[X_n - X_0] = C_1 C_2 t.$$

Choice of p and q

Recall $p + q = 1$, and we just showed $p - q = C_2\sqrt{\Delta t}$.

Solve this 2×2 linear system for p and q , we find:

$$p = \frac{1}{2}(1 + C_2\sqrt{\Delta t})$$

$$q = \frac{1}{2}(1 - C_2\sqrt{\Delta t})$$

Now, plugging p and q into our variance expression, we get:

$$\text{var}[X_n - X_0] = 4pqtC_1^2 = C_1^2t(1 - C_2^2\Delta t)$$

Finally, as $\Delta t \rightarrow 0$:

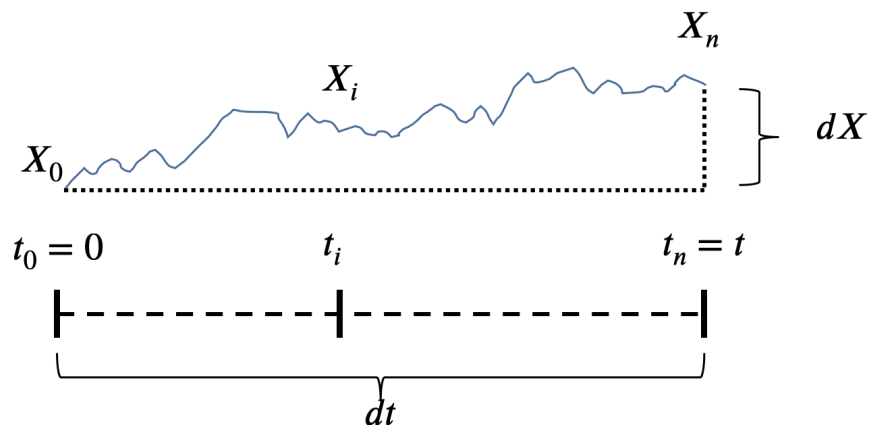
Both C_1 and C_2
are independent of
 Δt , so expectation
and variance are two.

$$E[X_n - X_0] = C_1C_2t$$

$$\text{var}[X_n - X_0] = C_1^2t$$

Relating Brownian motion and lattice walk

Now, assume that $X_n - X_0$ is very small, i.e., $X_n - X_0 \sim dX$, and likewise the time period $[0, t]$ is also very small, i.e., $t \sim dt$.



Making these substitutions, our random walk expressions become:

$$E[dX] = C_1 C_2 dt$$

and

$$\text{var}[dX] = C_1^2 dt$$

Relating Brownian motion and lattice walk

Let's compare the limiting behaviour of our lattice random walk results,

$$E[dX] = C_1 C_2 dt \quad \text{and} \quad \text{var}[dX] = C_1^2 dt$$

against that of the Brownian motion SDE ($dX = \alpha dt + \sigma dZ$):

$$E[dX] = \alpha dt \quad \text{and} \quad \text{var}[dX] = \sigma^2 dt$$

We can now choose C_1 and C_2 to force the first 2 “moments” (i.e., the mean & variance) to match each other so that the above results agree!

$$C_1 = \sigma$$

$$C_2 = \alpha/\sigma$$

Therefore, with these parameters we expect the discrete lattice walk to yield the same “solution” as the true SDE (in the limit of small time steps).

Relating Brownian motion and lattice walk

Definition: A *solution of an SDE* is an expression for the probability density of $X(t)$.

(Remember we said earlier there is no single answer/path, just a distribution of possible paths instead.)

It can indeed be proven that the probability density of the random walk on the lattice with $C_1 = \sigma$, $C_2 = \alpha/\sigma$, as $\Delta t \rightarrow 0$, converges to the solution of the SDE $dX = \alpha dt + \sigma dZ$.

So, lattice walk gives us a valid way to numerically (“discretely”) compute approximate solutions to Brownian motion with drift.

End of Lecture 5

Limit of Binomial Distribution

Note:

$$\lim_{\Delta t \rightarrow 0} E[X_n - X_0] = E[X(t) - X(0)] = \alpha t$$
$$\lim_{\Delta t \rightarrow 0} \text{var}[X_n - X_0] = \text{var}[X(t) - X(0)] = \sigma^2 t$$

Since $X_n - X_0$ (the lattice walk change) has a binomial distribution, and a binomial distribution converges to a normal distribution for large n ...

$$\lim_{\Delta t \rightarrow 0} (\text{binomial}) = \text{Normal}$$

then we have that $X(t) - X(0) \sim N(\alpha t, \sigma^2 t)$.

That is, the change in X under Brownian motion with drift is drawn from a particular normal distribution.

Examining a Special Case

Consider $\alpha = 0$, $\sigma = 1$. (i.e., zero drift, unit volatility). Then the SDE simplifies to:

$$dX = \alpha dt + \sigma dZ = dZ$$

and $Z(t) - Z(0) \sim N(0, t)$ according to the previous slide.

We can rewrite a random value drawn from $N(0, t)$ in terms of the standard normal distribution, $N(0, 1)$, using $\phi\sqrt{t}$, where $\phi \sim N(0, 1)$. So:

$$Z(t) - Z(0) = \phi\sqrt{t}$$

If $t \sim dt$ and $Z(0) = 0$, then:

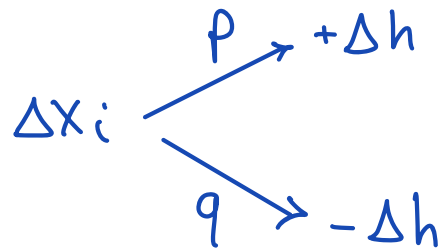
$$dZ = \phi\sqrt{dt} \text{ where } \phi \sim N(0, 1)$$

This give some intuition for the scaling of $dZ = \phi\sqrt{dt}$ in the SDE's random term.

(Z is sometimes called *standard* Brownian motion / a Wiener process.)

Summary

Computing a discrete random walk on a lattice
with parameters:



$$\Delta h = \sigma \sqrt{\Delta t},$$

$$p = \frac{1}{2} \left(1 + \frac{\alpha}{\sigma} \sqrt{\Delta t} \right),$$

$$q = \frac{1}{2} \left(1 - \frac{\alpha}{\sigma} \sqrt{\Delta t} \right)$$

converges as $\Delta t \rightarrow 0$ to the solution of the true SDE for Brownian motion with drift,

$$dX = \alpha dt + \sigma dZ$$

$$\text{with } dZ = \phi \sqrt{dt}$$

$$\text{and } \phi \sim N(0, 1).$$